NinjaRS in MadLoop

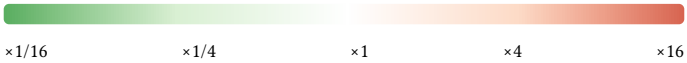
Fortran integration · scalar masters · measured benchmarks

2026-10-20 · 15:37 UTC · AMD EPYC 9754 128-Core Processor

Four reduction libraries, identical inputs. Each coloured card gives the median complete evaluation time in **µs per phase-space point**, with its multiplier against **original MadLoop + Ninja**. The new NinjaRS interface uses oneloopmaster scalar integrals. All candidate cards use the same MadLoop source and input stream.

QCD CORRECTIONS · LOOP_SM

Process	CutTools	Ninja	Collier	NinjaRS
$u\bar{u} \to Z$ 2 ⟶ 1 · 100 points qcd-1	µs / point **65.11** **×1.43** vs original Ninja reference 45.60 µs **ML init + warm 0.023 s** 7 retained repetitions MAD 0.17 µs / point	µs / point **38.89** **×0.85** vs original Ninja reference 45.60 µs **ML init + warm 0.018 s** 7 retained repetitions MAD 0.21 µs / point	µs / point **56.24** **×1.23** vs original Ninja reference 45.60 µs **ML init + warm 0.019 s** 7 retained repetitions MAD 0.30 µs / point	µs / point **38.60** **×0.85** vs original Ninja reference 45.60 µs **ML init + warm 0.009 s** 7 retained repetitions MAD 0.11 µs / point
$gg \to t\bar{t}$ 2 ⟶ 2 · 100 points qcd-2	µs / point **2194** **×1.53** vs original Ninja reference 1430 µs **ML init + warm 0.299 s** 7 retained repetitions MAD 33.73 µs / point	µs / point **1309** **×0.92** vs original Ninja reference 1430 µs **ML init + warm 0.176 s** 7 retained repetitions MAD 17.94 µs / point	µs / point **2227** **×1.56** vs original Ninja reference 1430 µs **ML init + warm 0.266 s** 7 retained repetitions MAD 36.46 µs / point	µs / point **1273** **×0.89** vs original Ninja reference 1430 µs **ML init + warm 0.176 s** 7 retained repetitions MAD 18.01 µs / point
$u\bar{u} \to Zgg$ 2 ⟶ 3 · 100 points qcd-3	µs / point — Timing pending 0 / 7 repetitions **ML init + warm — s** 0 retained repetitions No final timing inferred	µs / point — Timing pending 0 / 7 repetitions **ML init + warm — s** 0 retained repetitions No final timing inferred	µs / point — Timing pending 0 / 7 repetitions **ML init + warm — s** 0 retained repetitions No final timing inferred	µs / point — Timing pending 1 / 7 repetitions **ML init + warm — s** 1 retained repetitions No final timing inferred
$u\bar{u} \to ZZgg$ 2 ⟶ 4 · 100 points qcd-4	µs / point — Timing pending 0 / 7 repetitions **ML init + warm — s** 0 retained repetitions No final timing inferred	µs / point — Timing pending 0 / 7 repetitions **ML init + warm — s** 0 retained repetitions No final timing inferred	µs / point — Timing pending 0 / 7 repetitions **ML init + warm — s** 0 retained repetitions No final timing inferred	µs / point — Timing pending 0 / 7 repetitions **ML init + warm — s** 0 retained repetitions No final timing inferred



Logarithmic scale · lower is faster · white is parity · red is slower

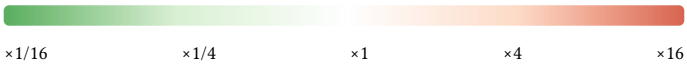
Snapshot: 16 / 32 timing cards complete; 4 / 8 correctness cases passed; 4 / 8 performance gates passed. Missing or incomplete observations remain pending. Measurements and validation are reported separately; an empty card is never treated as a speedup. These results describe the recorded implementation before the model-precision correction; the repaired generator requires a fresh validation series.

ELECTROWEAK CORRECTIONS · LOOP_QCD_QED_SM_GMU

Complete evaluation · electroweak controls

Real masses and zero widths; all generated virtual correction orders are retained. Full model names and MG5aMC process commands are listed with the reproducibility record.

Process	CutTools	Ninja	Collier	NinjaRS
$u\bar{d} \to W^+$ 2 ⟶ 1 · 100 points ew-1	µs / point **354.4** **×1.67** vs original Ninja reference 211.8 µs **ML init + warm 0.058 s** 7 retained repetitions MAD 2.47 µs / point	µs / point **199.8** **×0.94** vs original Ninja reference 211.8 µs **ML init + warm 0.044 s** 7 retained repetitions MAD 2.25 µs / point	µs / point **272.7** **×1.29** vs original Ninja reference 211.8 µs **ML init + warm 0.045 s** 7 retained repetitions MAD 3.78 µs / point	µs / point **195.8** **×0.92** vs original Ninja reference 211.8 µs **ML init + warm 0.041 s** 7 retained repetitions MAD 2.67 µs / point
$e^+e^- \to \mu^+\mu^-$ 2 ⟶ 2 · 100 points ew-2	µs / point **1413** **×1.72** vs original Ninja reference 821.3 µs **ML init + warm 0.202 s** 7 retained repetitions MAD 17.94 µs / point	µs / point **793.5** **×0.97** vs original Ninja reference 821.3 µs **ML init + warm 0.121 s** 7 retained repetitions MAD 4.13 µs / point	µs / point **1577** **×1.92** vs original Ninja reference 821.3 µs **ML init + warm 0.200 s** 7 retained repetitions MAD 8.27 µs / point	µs / point **852.3** **×1.04** vs original Ninja reference 821.3 µs **ML init + warm 0.121 s** 7 retained repetitions MAD 7.35 µs / point
$e^+e^- \to t\bar{t}\gamma$ 2 ⟶ 3 · 100 points ew-3	µs / point — Timing pending 0 / 7 repetitions **ML init + warm — s** 0 retained repetitions No final timing inferred	µs / point — Timing pending 0 / 7 repetitions **ML init + warm — s** 0 retained repetitions No final timing inferred	µs / point — Timing pending 0 / 7 repetitions **ML init + warm — s** 0 retained repetitions No final timing inferred	µs / point — Timing pending 1 / 7 repetitions **ML init + warm — s** 1 retained repetitions No final timing inferred
$e^+e^- \to \mu^+\mu^-\gamma\gamma$ 2 ⟶ 4 · 100 points ew-4	µs / point — Timing pending 0 / 7 repetitions **ML init + warm — s** 0 retained repetitions No final timing inferred	µs / point — Timing pending 0 / 7 repetitions **ML init + warm — s** 0 retained repetitions No final timing inferred	µs / point — Timing pending 0 / 7 repetitions **ML init + warm — s** 0 retained repetitions No final timing inferred	µs / point — Timing pending 0 / 7 repetitions **ML init + warm — s** 0 retained repetitions No final timing inferred



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Timing boundary. Timed calls include the MadLoop numerator, loop reduction, scalar integrals, assembly and normal stability checks or rescues, together with per-point scale/coupling updates and return-code counting. Output is written after the timed region. “ML init + warm” begins at the first MadLoop evaluation and includes helicity initialization and one complete additional point-stream pass. It excludes parameter-card/model initialization, input reading and allocation.

At least 7 complete repetitions are required before a card receives a timing. Calibration samples are excluded. Each repetition traverses the same 100 distinct input points enough times to target a one-second timed interval. Native backend caches remain part of normal execution.

VALIDATION · SAME INPUTS, EVERY COEFFICIENT

Numerical agreement

Process	Points	Orders	Checks	Max. error / bound	Status
$u\bar{u} \to Z$	100	2	4800	0.0000	passed
$u\bar{d} \to W^+$	100	2	4800	0.0000	passed
$gg \to t\bar{t}$	100	2	4800	0.0000	passed
$e^+e^- \to \mu^+\mu^-$	100	2	4800	0.0001	passed
$u\bar{u} \to Zgg$	—	—	—	—	pending
$e^+e^- \to t\bar{t}\gamma$	—	—	—	—	pending
$u\bar{u} \to ZZgg$	—	—	—	—	pending
$e^+e^- \to \mu^+\mu^-\gamma\gamma$	—	—	—	—	pending

Every point and squared coupling order contributes Born, finite, single-pole and double-pole values. The six pairings of CutTools, Ninja, Collier and NinjaRS are tested. "Orders" includes the total order slot. The reported maximum is the ratio below, where S is the largest absolute coefficient at that point across all orders and reducers. Agreement requires this ratio ≤ 1. Exactly zero pairs contribute zero.

$$\frac{|a-b|}{10^{-7}\max(|a|,|b|)+10^{-10}S}$$

Performance acceptance · NinjaRS / original Ninja

Process	Repetitions	Median ratio	95% interval	Decision
$u\bar{u} \to Z$	7	0.847	0.839 – 0.852	passed
$u\bar{d} \to W^+$	7	0.924	0.890 – 0.932	passed
$gg \to t\bar{t}$	7	0.890	0.889 – 0.900	passed
$e^+e^- \to \mu^+\mu^-$	7	1.038	1.020 – 1.044	passed
$u\bar{u} \to Zgg$	1	—	—	pending
$e^+e^- \to t\bar{t}\gamma$	1	—	—	pending
$u\bar{u} \to ZZgg$	0	—	—	pending
$e^+e^- \to \mu^+\mu^-\gamma\gamma$	0	—	—	pending

Acceptance threshold: ×1.05. The harness resamples paired run indices 10,000 times and forms a percentile 95% interval for the ratio of medians. A case passes when the upper bound is ≤ 1.05 it fails when the lower bound is > 1.05. An interval crossing the threshold is **unresolved**, not a pass. Each pair uses the same CPU, points and parameter card; measurement order alternates between repetitions.

Outstanding diagnostics. These are retained investigations, separate from accepted timing series.

Zgg: an exploratory seven-repeat series gave a NinjaRS/original-Ninja ratio of 2.269, above the 1.05 target. Concurrent task activity excludes this series from acceptance timing; the slowdown remains under investigation.

tt~gamma: tighter Collier integral accuracy removes the original status-410 flags. Corrections to generated MP complex constructors, pi and real casts pass analytic controls and all six backend comparisons at 100 fixed points in a fresh CLI export (largest tolerance fraction 0.000913). The original failed run remains retained; final recertification and explicit corrected-reference accounting are outstanding.

PREPARATION · SEPARATE WALL CLOCKS

Generation, compilation and initialization

Process	Original gen.	Original build	Candidate gen.	Candidate build	NinjaRS ML init + warm
$u\bar{u} \to Z$	87.86	0.98 ?	3.84	4.52 C	0.009
$u\bar{d} \to W^+$	109.75	0.93 ?	24.60	4.80 C	0.041
$gg \to t\bar{t}$	87.86	0.86 ?	7.67	5.61 C	0.176
$e^+e^- \to \mu^+\mu^-$	126.79	0.94 ?	48.03	5.01 C	0.121
$u\bar{u} \to Zgg$	92.62	0.89 ?	16.19	5.87 C	—
$e^+e^- \to t\bar{t}\gamma$	185.46	1.14 ?	107.86	6.22 C	—
$u\bar{u} \to ZZgg$	55.37	1.02 ?	54.27	9.83 C	—
$e^+e^- \to \mu^+\mu^-\gamma\gamma$	146.58	1.13 ?	153.55	10.17 C	—

All preparation values are seconds. Generation is the recorded CLI model import, process generation and standalone export invocation; it can include library work performed by export. Build is the retained `make -j4 check` invocation: **C** begins with no subprocess object files; **I** is incremental; **?** was not classified. A clean subprocess build does not include rebuilding external libraries. ML initialization plus warming is the median of the completed run preparations beginning at their first MadLoop call; parameter-card/model setup is excluded. Generation and compilation have no performance acceptance gate.

Measurement method

The process list is fixed in `cases.json`. For two-to-many scattering, the harness constructs deterministic RAMBO-derived on-shell momenta at 1000 GeV, including the generated external masses. For $2 \longrightarrow 1$, a physical on-shell resonance is produced with distinct three-dimensional boosts. Momentum conservation and masses are checked before measurements; each case uses a different deterministic seed.

The same retained text point stream and parameter-card SHA-256 hashes are required across reducers. All decay widths are set to zero. Each point sets the renormalization scale to its partonic centre-of-mass energy. Model parameters and couplings otherwise come from the generated card.

Each executable run selects one reduction library. The driver forces stability checks, allows normal rescue within that backend, and records every timed return code. The harness rejects initialization codes, unchecked estimates, failures and unexpected backend identities. It rebuilds helicity filters before each invocation. The baseline is the untouched original source revision recorded on the final page.

The process is pinned to one CPU with `taskset`; OpenMP and OpenBLAS thread counts are one. Generation may use four build jobs. Timing uses Fortran `SYSTEM_CLOCK`, and includes the driver's scale updates and histogram increment. Calibration, text input, numerical output and initialization are outside the warm interval. No separate scalar-master or numerator-stage timing is inferred from total runtime.

Limits of interpretation. Confidence intervals quantify the retained run-to-run samples on this host; they do not establish performance on other CPUs or workloads. A shared host can introduce noise. Finite random streams are a regression test, not a proof over all phase space. Pending correctness, rescue or timing results remain outstanding requirements.

STABILITY · ACTUAL BACKEND USE

Timed double- and higher-precision evaluations

Process	CutTools	Ninja	Collier	NinjaRS
$u\bar{u} \to Z$	107 100 DP 0.00 QP	179 200 DP 0.00 QP	125 300 DP 0.00 QP	177 100 DP 0.00 QP
$u\bar{d} \to W^+$	20 300 DP 0.00 QP	35 000 DP 0.00 QP	25 900 DP 0.00 QP	37 100 DP 0.00 QP
$gg \to t\bar{t}$	3500 DP 0.00 QP	5600 DP 0.00 QP	3500 DP 0.00 QP	5600 DP 0.00 QP
$e^+e^- \to \mu^+\mu^-$	5600 DP 0.00 QP	9100 DP 0.00 QP	4900 DP 0.00 QP	8400 DP 0.00 QP
$u\bar{u} \to Zgg$	Pending	Pending	Pending	100.0 DP 0.00 QP
$e^+e^- \to t\bar{t}\gamma$	Pending	Pending	Pending	100.0 DP 0.00 QP
$u\bar{u} \to ZZgg$	Pending	Pending	Pending	Pending
$e^+e^- \to \mu^+\mu^-\gamma\gamma$	Pending	Pending	Pending	Pending

Counts sum all retained timed repetitions, including partial schedules, and include repeated traversals of the point stream. They classify final successful evaluation status, rather than counting individual internal loop-reduction attempts. Full histograms are retained in `report-data.json` and the original measurement JSON. A pending entry means no accepted run is present.

Backward-compatible status interpretation

Backend	Precision	Hundreds / units digits
CutTools	DP / QP	$H = 2, U = 1$ / $H = 3, U = 9$
Ninja	DP / QP	$H = 2, U = 6$ / $H = 3, U = 8$
Collier	DP	$H = 2, U = 7$
NinjaRS	DP / QP	$H = 2, U = 8$ / $H = 3, U = 2$

The tens digit must identify a completed stability check (1 or 2). Old return integers are unchanged: the hundreds digit distinguishes Ninja QP from NinjaRS DP and PJFry++ DP from NinjaRS QP. Dedicated NinjaRS counters preserve legacy statistics interpretation.

Integration audit

The Fortran wrapper preserves MadLoop's momentum ordering, odd-rank sign reversal, thresholded invariants, Laurent ordering and normalization. Compiled recording-backend tests exercise both amplitude and interference paths, REAL16 input beyond binary64 precision, failures and nonfinite output, card parsing, old return codes, and cleanup with or without the installed module.

The bridge audit corrected the generic higher-precision auxiliary sampling scale to match Ninja's invariant/mass scale. Scalar tests cover spacelike massless and massive bubble orderings and reuse of coefficients, masses and scale. Higher-precision transport carries actual binary128 input into 256-bit arithmetic. Context cleanup joins the retained worker. See `audit.md` for scope, evidence and remaining limits.

REPRODUCIBILITY · PINNED SOURCES AND RETAINED INPUTS

Source and toolchain record

Component	Recorded revision / version
Original MadLoop	27147b3300f556a75661a886e92bdbe00558ee25
NinjaRS	0d0d7e7a8033431b60a8580731084742e0655cda
oneloopmaster	09b2c7497743db875ed446da95123175dd00b1c8
Symbolica	e679e8c129ce15ec50c46ffb4ff4b59aeb7813b8
Compilers	Rust 1.98.1 · gfortran 15.3.0
Report tools	Python 3.14.7 · Typst 0.15.1
Reference libraries	Ninja 1.2.1 · OneLOop 3.6 · Collier 1.2.9

Case	MG5aMC generate command	Point SHA-256 prefix
qcd-1	`u u~ > z [virt=QCD]`	bd8a00c86371
ew-1	`u d~ > w+ [virt=QED]`	5869f48fc054
qcd-2	`g g > t t~ [virt=QCD]`	6efb8cb1752f
ew-2	`e+ e- > mu+ mu- [virt=QED]`	ed6f3f830333
qcd-3	`u u~ > z g g [virt=QCD]`	b6121328eade
ew-3	`e+ e- > t t~ a [virt=QED]`	c030e6c68d53
qcd-4	`u u~ > z z g g [virt=QCD]`	d66b0c04ee3d
ew-4	`e+ e- > mu+ mu- a a [virt=QED]`	75a3765dca97

QCD cases use `loop_sm`; electroweak cases use `loop_qcd_qed_sm_Gmu`. Full dependency revisions, source archive and model hashes are in `provenance.json`. Full point/card hashes, per-run executable hashes when recorded, raw timings, Laurent coefficients, statuses and run logs remain with the benchmark inputs. `report-data.json` includes a SHA-256 manifest of the inputs used to build it.

Rebuild the measurements and report

```
source .ninjars-build/dev-env.sh
python tools/ninjars_benchmark.py generate --source both
python tools/ninjars_benchmark.py build --source both
python tools/ninjars_benchmark_workers.py measure --source both \
  --cases all --cpu 207 --repetitions 7 --clean-build --run-id core
python tools/ninjars_benchmark_workers.py promote --runs \
  docs/benchmarkfolder/experiments/persistent-workers/core
python tools/ninjars_report.py --compile
```

Run from the repository root after installing the pinned libraries and preparing the untouched baseline checkout. Choose an allowed quiet CPU; 207 is this run's choice. Refresh representative protocol validation as described in `persistent-workers.md`. Per-case selection uses `--cases qcd-1`. Failed and unresolved performance comparisons exit unsuccessfully and retain their JSON evidence. The report generator rejects stale summaries and mismatched input streams instead of silently combining them.

Style attribution. Typography, margins, navy headings, pale notes, striped tables, provider cards and the logarithmic green–white–red ratio palette follow MadGraphTeam/pyamplicol's one-loop benchmark report. Its physics results and performance claims are not reused here.